



1) 6, 8, 9, 10

$$\frac{2^6}{2^4 \cdot 2^{-1}} = 2^{6-4-(-1)} = 2^3 = 8$$

$$\frac{6^{12}}{2^9 \cdot 3^{11}} = \frac{2^{12} \cdot 3^{12}}{2^9 \cdot 3^{11}} = 2^3 \cdot 3 = \underline{\underline{24}}$$

$$\begin{aligned} g) \quad 7^{\frac{4}{9}} \cdot 49^{\frac{5}{18}} &= 7^{\frac{4}{9}} \cdot \left(7^{\frac{5}{9}}\right)^2 = \\ 7^{\frac{4}{9} + \frac{10}{9}} &= 7^{\frac{14}{9}} = 7^1 = 7 \end{aligned}$$

$$\begin{aligned} &(-10)^3 + (-10)^2 + (-10)^0 = \\ -1000 + 100 + 1 &= \underline{\underline{-899}} \quad (+) \end{aligned}$$

$$x^2 + 12 = 7x$$

$$x = 2$$

$$x = 4$$

$$49 - 48 = 1$$

$$x^2 - 7x + 12 = 0$$

$$a = 1$$

$$b = -7$$

$$c = 12$$

$$\frac{-(-7) \pm \sqrt{49 - 4 \cdot 12}}{2 \cdot 1} =$$

$$\frac{7 \pm 1}{2} \rightarrow \begin{matrix} 4 \\ 3 \end{matrix}$$

$$\frac{-b \pm \sqrt{D}}{2a}$$

$$x^2 - 17x + 72 = 0$$

Меньше!

$$b = -17$$

$$\sqrt{1} = 1 = \sqrt{1}$$

$$\frac{-(-17) \pm 1}{2}$$

$$D = 17^2 - 4 \cdot 72$$

$$289 - 288 = \underline{\underline{1}}$$

$$= \frac{17 \pm 1}{2} \rightarrow \begin{matrix} 8 \\ 9 \end{matrix}$$

$$(2x+4)^2 = (2x-1)^2$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$(2x)^2 + 2 \cdot 2x \cdot 4 + 4^2$$

$$\underline{4x^2} + \underline{28x} + 49 = 4x^2 - 2 \cdot 2x \cdot 1 + 1$$

$$4x^2 + 28x + 49 = 4x^2 - 4x + 1$$

$$\cancel{4x^2 + 28x + 49} - \cancel{4x^2 - 4x + 1} = 0$$

$$32x + 48 = 0$$

$$32x = -48$$

$$x = \frac{-48}{32}$$

$$x^2 + 9 = (x+9)^2$$

$$x^2 + 2 \cdot x \cdot 9 + 9^2$$

$$x^2 + 9 = x^2 + 18x + 81$$

$$\cancel{x^2} + 18x + 81 - \cancel{x^2} - 9 = 0$$

$$18x + 72 = 0$$

$$18x = -72$$

$$x = \frac{-72}{18}$$

$$(x-6)^2 = -24x$$

$$(a^2 - 2ab + b^2)$$

$$a = x$$

$$b = 6$$

$$x^2 - 2 \cdot 6 \cdot x + 36 = -24x$$

$$x^2 - 12x + 36 = -24x$$

$$x^2 - 12x + 36 + 24x = 0$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(3x+2)^2 = 9x^2 + 2 \cdot 3x \cdot 2 + 4$$

$$9x^2 + 12x + 4$$

$$x^2 + 12x + 36 = 0$$

$$144 - 36 \cdot 4 = 0$$

$$\frac{-12}{2} = (-6)$$

$$(5x-3)^2 = (5x+13)^2$$

$$25x^2 - 2 \cdot 5x \cdot 3 + 9 = 25x^2 + 2 \cdot 5x \cdot 13 + 169$$

$$25x^2 - 30x + 9 = 25x^2 + 130x + 169$$

$$\cancel{25x^2} + \underline{\underline{130x}} + \underline{\underline{169}} - \cancel{25x^2} - \underline{\underline{130x}} - \underline{\underline{9}} = 0$$

$$160x + 160 = 0$$

$$160x = -160$$

$$\boxed{x = -1}$$

Д/З Как bubbles эти формулы